Amber Lo

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EDUCATION

Carnegie Mellon University | Pittsburgh, PA — May 2027
Bachelor of Science in Chemical Engineering, Additional Major in Biomedical Engineering
EPFL – Swiss Federal Institute of Technology Lausanne | Lausanne, VD | Study Abroad — February 2026 - May 2026

SKILLS

Biomedical Engineering: Plasmid cloning, retroviral vector production, primary T-cell engineering, mammalian cell culture, flow cytometry, FlowJo, Miltenyi MACS enrichment, PCR, mice tissue/organ processing
Chemical Engineering & Fabrication: Liquid-liquid extraction, soft lithography (microfluidic device fabrication), CAD (Onshape), HPLC, GC, UV-Vis
Data & Programming: Python (SciPy Optimization), FlowJo, GraphPad Prism, MATLAB, COMSOL, Aspen Plus, Excel, Adobe Illustrator, PowerPoint

RESEARCH EXPERIENCE

EPFL Undergraduate Researcher, Dr. Tang Lab — June 2026 – August 2026

- Developed an in vitro T-cell exhaustion model by creating an antibody staining panel for flow cytometry analysis
- Engineered primary murine T-cells through retroviral transduction and evaluated receptor signaling and antitumor function using STAT and ERK signal staining and luciferase-based killing assays
- Cloned plasmids and produced concentrated retroviral vectors for T-cell engineering Enriched IL13R1 and IL13R2 from 40% transduction efficiency to 95% using magnetic bead enrichment (Miltenyi MACS)
- Analyzed experimental data using FlowJo and GraphPad Prism, generated hypotheses, proposed follow-up studies, and presented findings to the laboratory

CMU Undergraduate Researcher, Dr. Sevenler Lab — January 2025 – January 2026

- Engineered Jurkat and NK cells via viscoelastic mechanoporation achieving 92% and 49% transfection efficiency, respectively
- Designed and optimized erythrocyte transfection protocols in collaboration with the VELVET team, achieving 79.6% delivery efficiency
- Manufactured microfluidic devices via soft lithography for CRISPR-Cas9 delivery
- Trained new lab members in mammalian cell culture techniques, flow cytometry characterization, and Computer-Aided Design (Onshape)
- Delivered a poster presentation at the BERG Symposium communicating research findings

PROJECT EXPERIENCE

Alkylation Process Optimization | Numerical Methods — October 2025

- In teams of 3, developed a computational optimization model to maximize process profitability for an alkylation plant by minimizing operational costs using Python (scipy.optimize)
- Applied numerical optimization techniques to evaluate process tradeoffs and identify optimal operating conditions
- Presented modeling approach and optimization results to a class audience and the Department Head

Nicotine Concentration Analysis | Introduction to Chemical Analysis — May 2025

- Quantified active compound concentration across nicotine patches and Zyns using UV-Vis spectroscopy, HPLC, and gas chromatography (GC)
- Performed sample preparation and purification using liquid–liquid extraction and rotary evaporation to isolate analytes from complex matrices
- Evaluated compound delivery efficiency using comparative analytical measurements and data interpretation

ADDITIONAL EXPERIENCE

Lead Tutor | PLUS Tutors — January 2024 – Present

- Led tutoring sessions supporting middle school students in mathematics, contributing to mastery of 148 skills across 28 students in one semester

LEADERSHIP AND ACTIVITIES

Treasurer | ChemE Cube — August 2025 – February 2026

- Secured funding through successful grant proposals and CMU Giving Day, increasing club budget by 58% and enabling four additional students to attend the AIChE Conference, supporting prototyping initiatives and student-led engineering projects

Sophomore Representative | American Institute of Chemical Engineers (AIChE) — August 2024 – May 2025

- Founded and organized “ChemE Day,” a department-wide event engaging 350+ students, faculty, and staff to strengthen community involvement

Summer Undergraduate Researcher Fellowship Recipient — 2025
Member | Hong Kong Student Association — January 2024 - Present
Member | Kappa Phi Lamba — September 2023 - Present